

TEACHER'S NAME: _____

STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

YEAR 12 HALF YEARLY EXAMINATION

2007

MATHEMATICS EXTENSION 2

*Time allowed - Three hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Begin each Question on a fresh page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

QUESTION 1 (15 Marks)**Marks**

- (a) The complex number z is given by $z = \frac{2+i}{2-i} - 2$. Find: 3
- (i) $\bar{z}$
(ii) $\text{Im}(z)$
(iii) $|z|$
- (b) Draw neat labelled sketches to show each of the following graphs on the Argand diagram. 4
- (i) $|z - iz| = 2$
(ii) $|z - 1 - 2i| \leq 3$ and $|z + i| \leq |z - i|$
- (c) Prove $|z_1 + z_2| \leq |z_1| + |z_2|$ and explain when the equality holds. 3
- (d) Solve for $z = x + iy$ the equation $z\bar{z} + 2z = \frac{1}{4} + i$. 3
- (e) Use De Moivre's Theorem to express $\cos 5\theta$ in terms of $\cos \theta$ and $\sin \theta$. 2

QUESTION 2 (15 Marks)

- (a) Solve $z^2 - 5iz + 2 = 0$ 2
- (b) Given $z = 3 - 3\sqrt{3}i$, 3
- (i) Find z in mod-arg form
(ii) Find z^4 in the form $x + iy$.
- (c) Find the square roots of $4 + 3i$. 3
- (d) Describe in words and sketch the locus of z if $\text{Arg}\left(\frac{z-1}{z-2i}\right) = \frac{\pi}{2}$. 3
- (e) Solve $z^5 = 1$ over the complex field and plot the solutions on an Argand diagram. 3
- (f) Find i^{2009} . 1

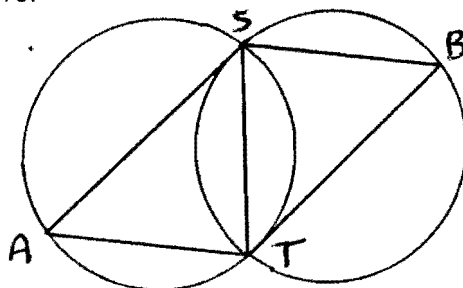
QUESTION 3 (15 Marks)

Marks

- (a) The ellipse E is defined by E: $\frac{x^2}{9} + \frac{y^2}{4} = 1$
- (i) For E write down its eccentricity, co-ordinates of its foci and equations of its directrices, and sketch the curve, showing its features. 4
- (ii) Show that the equation of the normal at P ($3\cos\theta, 2\sin\theta$) is $\frac{3x}{\cos\theta} - \frac{2y}{\sin\theta} = 5$. 3
- (iii) If the normal at P cuts the major axis at G and the minor axis at H, then find M, the midpoint of GH. 3
- (b)
- (i) Show that for an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the sum of the focal distances is $2a$. 2
- (ii) Hence find the equation of the locus of z such that $|z + 2i| + |z - 2i| = 6$ 3

QUESTION 4 (15 Marks)

- (a)
- (i) Show that in the first quadrant the ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $4x^2 - y^2 = 4$ intersect at right angles. 4
- (ii) Find the equation of the one circle through the points of intersection of the two conics in all four quadrants. 2
- (b) Find the equation(s) of the normal to the curve $xy(x + y) + 2 = 0$ at the point(s) where the gradient of the tangent is -1. 5
- (c) AS and TB are tangents. Prove:
- (i) $\triangle SAT \sim \triangle BTS$ 2
- (ii) $AT \times BS = ST^2$ 1
- (iii) $SB \parallel AT$ 1



QUESTION 5 (15 Marks)**Marks**

- (a) The rectangular hyperbola H has equation $x^2 - y^2 = 2$. Write down its eccentricity, the co-ordinates of its foci S and S' , and vertices A and A' , the equations of its directrices and asymptotes. 4
- (b)
- (i) Show the condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$. 3
- (ii) Show that the pair of tangents drawn from $(5,0)$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to each other. 2
- (c)
- (i) Prove that the equation of the tangent to $xy = c^2$ at the point $P(cp, \frac{c}{p})$ is $x + p^2y = 2cp$. 2
- (ii) If the tangent at P passes through the foot of the ordinate of $Q(cq, \frac{c}{q})$ show that $q = 2p$. 2
- (iii) Hence prove that the locus of the midpoint of PQ is $xy = \frac{9c^2}{8}$. 2

QUESTION 6 (15 Marks)

- (a) Sketch the graph of $y = (2x + 5)(x + 1)$ showing clearly the intercepts on the axes and the co-ordinates of any turning points. 2
- (b) Use the graph in part (a) to sketch the graph of $y = \frac{1}{(2x + 5)(x + 1)}$ showing clearly the intercepts on the axes, the co-ordinates of any turning points and the equations of any asymptotes. 4
- (c) Use the graph in part (a) to sketch the graph of $y = \ln[(2x + 5)(x + 1)]$ showing the y intercept and the equations of any asymptotes. 4
- (d) Find the equation of the tangent to the curve $y = \ln[(2x + 5)(x + 1)]$ at $x = 0$. 3
- (e) Hence find the values of k such that exactly one of the solutions of the equation $\ln[(2x + 5)(x + 1)] = kx + \ln 5$ is a positive number. 2

QUESTION 7 (15 Marks)

Marks

- (a) Sketch the graph of $f(x) = \frac{3x^2 - 7}{(x-2)(x+3)}$, given that

$$f'(x) = \frac{(3x-1)(x-7)}{(x-2)^2(x+3)^2} \quad 5$$

- (b) Hence draw neat half-page sketches of:

(i) $y = \sqrt{f(x)}$ 2

(ii) $y = (f(x))^2$ 2

(iii) $y = f(|x|)$ 2

- (b) (i) Draw the graph of $y = |2x+1| - |x-2|$ in the domain $-6 \leq x \leq 3$ 2

(ii) Use the graph to solve $|2x+1| - |x-2| \leq 2$ 2

QUESTION 8 (15 Marks)

- (a) For what values of a does $\int_1^a (x + \frac{1}{2}) dx = 2 \int_0^{\frac{\pi}{4}} \sec^2 x dx$? 3

(b)

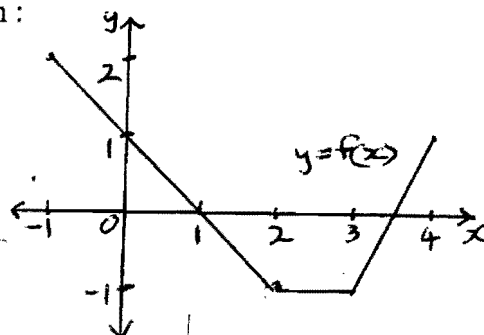
- (i) Write an expression for $\cos(A+B)$ and $\cos(A-B)$. Hence express $2\sin A \sin B$ as a difference of two functions. 2

- (ii) Given that $\sin x \sin 3x + \sin x \sin 5x + \sin x \sin 7x = \sin ax \sin bx$ then find the values of a and b . 3

- (c) Given p and q are roots of $ax^2 + bx + c = 0$, show, without solving for p or

q , that $\frac{q}{ap+b} + \frac{p}{aq+b} = \frac{-2}{a}$. 4

- (d) For the graph of $y = f(x)$ shown :



- (i) Find $\int_0^4 f(x) dx$ 2

- (ii) For what value of a , where $0 \leq a \leq 4$, does $\int_0^a f(x) dx$ have the smallest value? 1

$$\begin{aligned} Q1(a) \quad z &= \frac{2+i}{2-i} - 2 = \frac{2+i}{2-i} \times \frac{2+i}{2+i} - 2 \\ &= \frac{4+4i-1}{5} - 2 = \frac{3}{5} + \frac{4i}{5} - 2 \\ &= -\frac{7}{5} + \frac{4i}{5} \end{aligned}$$

$$(i) \quad \bar{z} = -\frac{7}{5} - \frac{4i}{5} \quad (ii) \quad 4\operatorname{Im} z = \frac{4}{5}$$

$$(iii) \quad |z| = \sqrt{\frac{49}{25} + \frac{16}{25}} = \frac{\sqrt{65}}{5}$$

$$(b)(i) \quad |z - iz| = 2$$

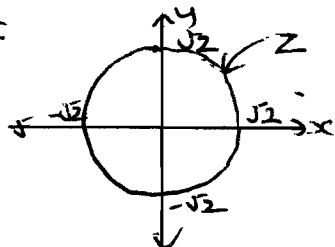
$$|(x+iy) - i(x+iy)| = 2$$

$$|x+iy + i(y-ix)| = 2$$

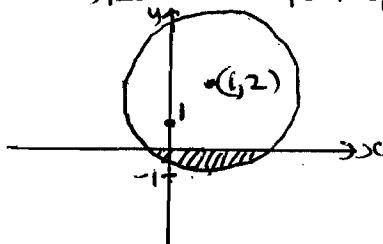
$$(x+y)^2 + (y-x)^2 = 4$$

$$x^2 + 2xy + y^2 + y^2 - 2yx + x^2 = 4$$

$$\therefore x^2 + y^2 = 2$$



$$(ii) \quad |z - (1+2i)| \leq 3 \text{ and } |z+i| \leq |z-i|$$



(c) In $\triangle OAC$

$$OA = |z_1|, AC = OB = |z_2|$$

$$OC = |z_1 + z_2|$$

In any $\triangle$, $OC < OA + AC$

$$\therefore |z_1 + z_2| < |z_1| + |z_2|$$

Equality holds when $\arg z_1 = \arg z_2$

or O, A, B are collinear.

$$(d) \quad x^2 + y^2 + 2x + 2iy = \frac{1}{4} + i$$

$$\therefore 2y = 1 \quad \therefore y = \frac{1}{2}$$

$$x^2 + \frac{1}{4} + 2x = \frac{1}{4}$$

$$x(x+2) = 0 \quad \therefore x = 0 \text{ or } -2$$

$$\therefore z = \frac{1}{2}i \text{ or } z = -2 + \frac{1}{2}i$$

$$(e) \quad z^5 = (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{or } z^5 = \cos^5 \theta + 5 \cos^4 \theta i \sin \theta + 10 \cos^3 \theta i^2 \sin^2 \theta + 10 \cos^2 \theta i^3 \sin^3 \theta + 5 \cos \theta i^4 \sin^4 \theta + i^5 \sin^5 \theta$$

$$= (\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta)$$

$$+ i(5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta)$$

Equating real parts give

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\begin{aligned} Q2(a) \quad z &= \frac{5i \pm \sqrt{25-8}}{2} = \frac{5i \pm i\sqrt{17}}{2} \\ &= \frac{(5 \pm \sqrt{17})i}{2} \end{aligned}$$

$$(b)(i) \quad r = \sqrt{9+27} = \sqrt{36} = 6$$

$$\tan(\arg z) = -\frac{3\sqrt{3}}{3} = -\sqrt{3} \quad \therefore \arg z = -\frac{\pi}{3}$$

$$\therefore z = 6 \left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$$

$$(ii) \quad z^4 = 6^4 \left[\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) \right]$$

$$= 1296 \left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3} \right)$$

$$= 1296 \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right)$$

$$= -648 + 648i\sqrt{3}$$

$$(c) \text{ Let } a+ib = \sqrt{4+3i}$$

$$\therefore a^2 - b^2 + 2iab = 4 + 3i$$

$$\therefore a^2 - b^2 = 4 \text{ and } 2ab = 3 \quad \therefore b = \frac{3}{2a}$$

$$\therefore a^2 - \frac{9}{4a^2} = 4 \text{ or } 4a^4 - 9a^2 - 4 = 0$$

$$(2a^2 - 9)(2a^2 + 1) = 0$$

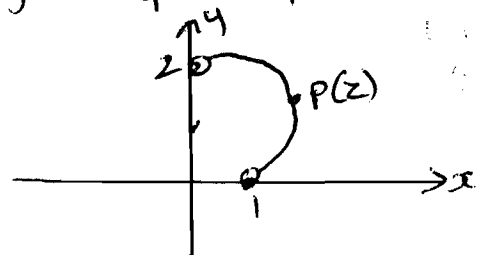
$$a^2 = \frac{9}{2} \text{ or } a^2 = -\frac{1}{2} \quad \therefore \text{No soln.}$$

$$a = \pm \frac{3}{\sqrt{2}}, \quad b = \pm \frac{3}{2} \times \frac{\sqrt{2}}{3} = \pm \frac{1}{\sqrt{2}}$$

$$\therefore \sqrt{4+3i} = \pm \left(\frac{3}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$$

$$(d) \quad \arg(z-1) - \arg(z-2i) = \frac{\pi}{2}$$

Locus is a semi-circle with diameter endpoints $(0, 2)$ and $(1, 0)$ but not including endpoints.



$$(e) \quad z^5 = (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\therefore \cos 5\theta = \cos 2k\pi, \text{ for } k=0, 1, 2, 3, 4$$

$$\therefore \theta = \frac{2k\pi}{5}, \quad k=0, 1, 2, 3, 4$$

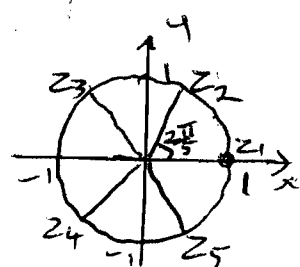
$$\therefore z_1 = 1$$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$z_4 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$



$$(f) \quad i^4 = 1$$

$$\therefore i^{2008} = 1$$

$$\therefore i^{2009} = i$$

Q3 (a) $a=3, b=2$

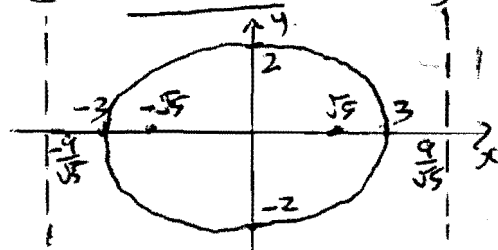
(i) $b^2 = a^2(1-e^2)$

$4 = 9(1-e^2)$

$1-e^2 = \frac{4}{9}, e^2 = \frac{5}{9}, e = \frac{\sqrt{5}}{3}$

Foci $= (\pm ae, 0) = (\pm\sqrt{5}, 0)$

Dir. are $x = \pm \frac{a}{e} \therefore x = \pm \frac{9}{\sqrt{5}}$ or $x = \pm \frac{9\sqrt{5}}{5}$



(ii) Diff. Imp.

$\frac{2x}{9} + \frac{2y}{4} y' = 0 \therefore y' = -\frac{2x}{9} \times \frac{2}{y} = -\frac{4x}{9y}$

$m_{Tang} = -\frac{12 \cos \theta}{18 \sin \theta} = -\frac{2 \cos \theta}{3 \sin \theta}$

$\therefore m_{Norm} = \frac{3 \sin \theta}{2 \cos \theta}$

Eqn of normal is

$y - 2 \sin \theta = \frac{3 \sin \theta}{2 \cos \theta} (x - 3 \cos \theta)$

$\frac{2y}{\sin \theta} - 4 = \frac{3x}{\cos \theta} - 9$

$\frac{3x}{\cos \theta} - \frac{2y}{\sin \theta} = 5$

(iii) At G, $y=0 \therefore x = \frac{5 \cos \theta}{3}$

At H, $x=0 \therefore y = -\frac{5 \sin \theta}{2}$

$\therefore M = (\frac{5}{6} \cos \theta, -\frac{5}{4} \sin \theta)$

(b)(i) $PS = e PM$ where M is on d

$= e(\frac{a}{e} - x_1) = a - ex_1$

$PS' = e PM'$ where M' is on d'

$= e(x_1 - (-\frac{a}{e})) = ex_1 + a$

$PS + PS' = a - ex_1 + ex_1 + a = 2a$

(ii) $PS + PS' = 2b$ where $S = (0, 2)$

$S' = (0, -2), b=3$

$\therefore be = 2 \therefore e = \frac{2}{3}$

$a^2 = b^2(1-e^2)$

$= 9(1-\frac{4}{9})$

$= 9 \times \frac{5}{9} = 5$

$\therefore a = \sqrt{5}$

$\therefore$ Eqn is

$\frac{x^2}{5} + \frac{y^2}{9} = 1$

Q4 (a) (i) For $4x^2 + 9y^2 = 36$, diff implicitly

$8x + 18y y' = 0 \therefore y' = -\frac{4x}{9y}$

For $4x^2 - y^2 = 4$, diff. imp.

$8x - 2yy' = 0 \therefore y' = \frac{4x}{y}$

For $y^2 = 4x^2 - 4$ sub in $4x^2 + 9y^2 = 36$

$4x^2 + 36x^2 - 36 = 36$

$40x^2 = 72$

$x^2 = \frac{9}{5} \therefore x = \frac{3}{\sqrt{5}} \text{ or } -\frac{3}{\sqrt{5}}$

$\therefore y^2 = \frac{36}{5} - 4 = \frac{16}{5} \therefore y = \pm \frac{4}{\sqrt{5}}$

$\therefore y'_1 \times y'_2 = -\frac{4}{9} \times \frac{3}{\sqrt{5}} \times \frac{\sqrt{5}}{4} \times \frac{4}{3} \times \frac{\sqrt{5}}{4} = -1$

$\therefore$ Intersect at right angles

(ii) By symmetry, pts of intersection are $(\pm \frac{3}{\sqrt{5}}, \pm \frac{4}{\sqrt{5}})$. Centre $= (0, 0)$

$OP = \sqrt{\frac{9}{5} + \frac{16}{5}} = \sqrt{5}$

$\therefore$ Eqn is $x^2 + y^2 = 5$

(b) $x^2y + xy^2 + 2 = 0$

Diff. impl.

$y \cdot 2x + x^2 y' + y^2 \cdot 1 + x \cdot 2y y' = 0$

$y'(x^2 + 2xy) = -y^2 - 2xy$

$y' = \frac{-y^2 - 2xy}{x^2 + 2xy}$

If $m_{Tang} = -1, -y^2 - 2xy = -x^2 - 2xy$
 $\therefore x^2 = y^2$
 $x = \pm y$

If $x = y, x^3 + x^3 + 2 = 0 \therefore x = -1$

If $-x = y, -x^3 + x^3 + 2 = 0 \therefore$ No soln.

$m_N = 1$

$\therefore$ Eqn of normal is $y + 1 = 1(x + 1)$
 $y = x$

(c) (i) In $\Delta SAT, \Delta BTS$

$\angle SAT = \angle BTS$ ($\angle$ bet tang. & chord)
 $= \angle$ in alt. seg.

$\angle AST = \angle SBT$ ($\angle$ bet tang. & chord)
 $= \angle$ in alt. seg.

$\therefore \angle ATS = \angle BST$ (3rd $\angle$ s in Δ s)

$\therefore \Delta SAT \parallel \Delta BTS$ (AAA)

(ii) $\frac{ST}{AT} = \frac{BS}{ST}$ (Match sides in sim. Δ s are in same ratio)

(iii) $\angle ATS = \angle BST$ (Eq. $\angle$ s in sim. Δ s)

$\therefore SB \parallel AT$ (Alt. $\angle$ s equal)

Q5(a) $\frac{x^2}{2} - \frac{y^2}{2} = 1 \therefore a = \sqrt{2}, b = \sqrt{2}$

$b^2 = a^2(e^2 - 1) \therefore e^2 - 1 = 1 \therefore e^2 = 2 \therefore e = \sqrt{2}$

$\therefore \text{Foci} = (\pm ae, 0) = (\pm 2, 0)$

Vertices $= (\pm a, 0) = (\pm \sqrt{2}, 0)$

Directrices are $x = \pm \frac{a}{e} \therefore x = \pm 1$

Asymptotes are $y = \pm x$

(b)(i) solve $y = mx + c$ with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore \frac{x^2}{a^2} + \frac{m^2x^2 + 2mxc + c^2}{b^2} = 1$

$b^2x^2 + a^2m^2x^2 + 2a^2mxc + a^2c^2 = a^2b^2$

$(b^2 + a^2m^2)x^2 + 2a^2mxc + a^2c^2 - a^2b^2 = 0$

$\therefore$ tangent $\Delta = 0$

$\therefore 4a^4m^2c^2 = 4(b^2 + a^2m^2)a^2(c^2 - b^2)$

$a^2m^2c^2 = b^2c^2 - b^4 + a^2m^2c^2 - a^2b^2m^2$

$b^4 + a^2b^2m^2 = b^2c^2$

$\therefore c^2 = a^2m^2 + b^2$

(ii) Since $y = mx + c$ then

$y = mx \pm \sqrt{16m^2 + 9}$

Sub in $(5, 0)$

$\therefore 0 = 5m \pm \sqrt{16m^2 + 9}$

$16m^2 + 9 = 25m^2$

$9m^2 = 9$

$m^2 = 1$

$m = \pm 1$

Tangents with gradients 1 and -1

$\therefore$ Perpendicular.

(c)(i) $xy = c^2 \therefore y = \frac{c^2}{x}$

$\therefore y' = -c^2x^{-2} = -\frac{c^2}{x^2}$

At P $(c, \frac{c}{p})$ $m_T = -\frac{c^2}{c^2p^2} = -\frac{1}{p^2}$

$\therefore$ Eqn of tangent is

$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$p^2y - cp = -x + cp$

$p^2y + x = 2cp$

(ii) Tangent passes thru $(cq, 0)$

$\therefore 0 + cq = 2cp \therefore q = 2p$

(iii) $M_{PQ} = (\frac{c(p+q)}{2}, \frac{\frac{c}{p} + \frac{c}{q}}{2})$

let $q = 2p$

$\therefore M_{PQ} = (\frac{3pc}{2}, \frac{c}{2} \cdot \frac{3}{2p})$

$\therefore$ If $x = \frac{3cp}{2}$ and $y = \frac{3c}{4p}$

then $xy = \frac{9c^2}{8}$

6(a) $y = (2x+5)(x+1)$

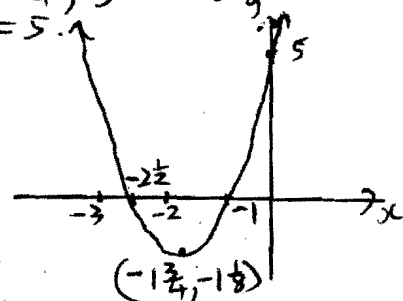
$= 0$ at $x = -2\frac{1}{2}, x = -1$

Vertex at $x = -1\frac{3}{4}, y = -\frac{9}{8}$

When $x = 0, y = 5$

int = 1

turning pt =

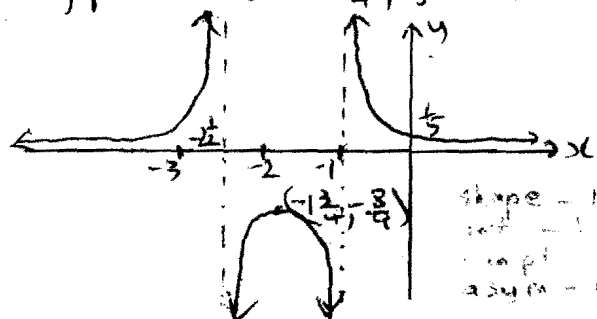


(b) $y = \frac{1}{(2x+5)(x+1)}$

Asym. at $x = -2\frac{1}{2}, x = -1$

y-int at $y = \frac{1}{5}$

Turning pt at $x = -1\frac{3}{4}, y = -\frac{8}{9}$

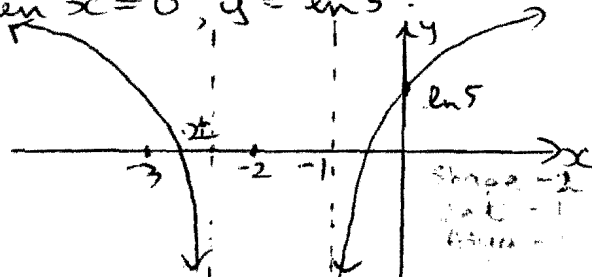


(c) $y = \ln[(2x+5)(x+1)]$

Exists when $y = (2x+5)(x+1)$ is pos.

$\therefore$ For $x < -2\frac{1}{2}$ and $x > -1$

When $x = 0, y = \ln 5$



(d) $y' = \frac{(x+1)2 + (2x+5)1}{(2x+5)(x+1)} = \frac{4x+7}{(2x+5)(x+1)}$

When $x = 0, m_T = \frac{7}{5}$ and $y = \ln 5$

$\therefore$ Eqn. is $y = \frac{7}{5}x + \ln 5$

(e) If $k = \frac{7}{5}$, tangent touches at

$x = 0$ and crosses for $x < -2\frac{1}{2}$

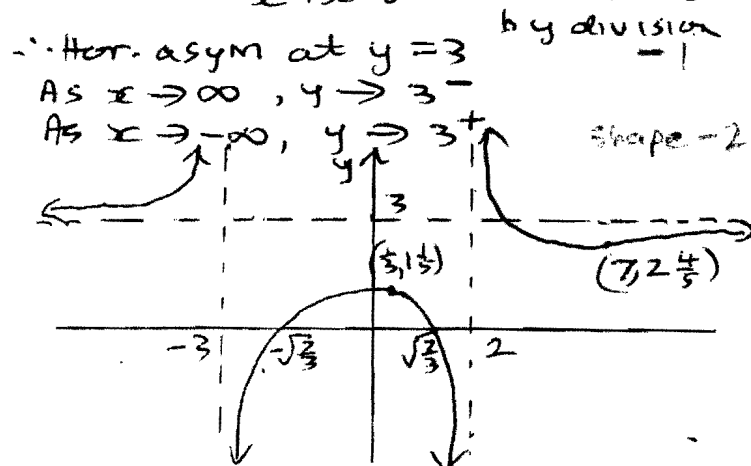
If $0 < k < \frac{7}{5}$, line crosses

for $x < -2\frac{1}{2}$, for $x = 0$ and

for $x > 0$

$\therefore$ One positive soln if $0 < k < \frac{7}{5}$

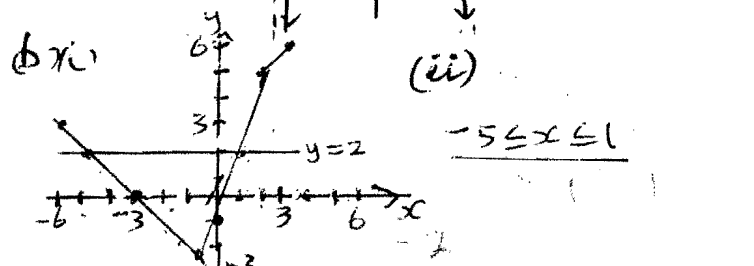
27. (a) Asymptotes at $x = -3, x = 2$ -
 St. pts when $f'(x) = 0$
 $\therefore x = \frac{1}{3}, y = 1\frac{1}{3}$ and $x = 7, y = 2\frac{4}{5}$ -
 When $-3 < x < \frac{1}{3}$, $f'(x) > 0$ $\therefore$ Max at $x = \frac{1}{3}$
 When $\frac{1}{3} < x < 2$, $f'(x) < 0$
 When $2 < x < 7$, $f'(x) < 0$ $\therefore$ Min at $x = 7$
 When $x > 7$, $f'(x) > 0$
 For $f(x) = \frac{3x^2 - 7}{x^2 + x - 6} = 3 + \frac{11 - 3x}{x^2 + x - 6}$



(b) (i) $y = \sqrt{f(x)}$ Only exists for $f(x) \geq 0$
 Asym at $x = -3, x = 2$. Hor. Asym, $y = \sqrt{3}$
 $y' = \frac{f'(x)}{2\sqrt{f(x)}}$ St. pts when $f'(x) = 0$
 Der. does not exist when $f(x) = 0$ Shape - 1
 Asym - 1

(ii) $y = (f(x))^2 \therefore y' = 2(f(x))f'(x) = 0$
 when $f(x) = 0$ and $f'(x) = 0$
 i.e. when $x = \frac{1}{3}, 7, \pm\sqrt{3}$ Shape - 1
 Hor. asymp $y = 9$ Asym - 1

(iii) $y = f(|x|)$. y' is undef at $x = 0$



Q8 (a) $\int_0^a (x + \frac{1}{2}) dx = 2 \int_0^{\frac{\pi}{4}} \sec^2 x dx$ - 15
 $[\frac{x^2}{2} + \frac{x}{2}]_0^a = 2[\tan x]_0^{\frac{\pi}{4}}$ Int. - 1
 Int. - 1
 $(\frac{a^2}{2} + \frac{a}{2}) - (\frac{1}{2} + \frac{1}{2}) = 2 - 0$
 $\frac{a^2}{2} + \frac{a}{2} - 3 = 0$
 $a^2 + a - 6 = 0$
 $(a + 3)(a - 2) = 0$

$\therefore a = -3 \text{ or } 2$ - 1

(b) $\cos(A+B) = \cos A \cos B - \sin A \sin B$ --- (1)
 $\cos(A-B) = \cos A \cos B + \sin A \sin B$ --- (2)
 $\therefore (2) - (1)$ gives
 $\cos(A-B) - \cos(A+B) = 2 \sin A \sin B$ - 1

(ii) $\sin x \sin 3x + \sin x \sin 5x + \sin x \sin 7x$
 $= \frac{1}{2}(\cos 2x - \cos 4x + \cos 4x - \cos 6x + \cos 6x - \cos 8x)$
 $= \frac{1}{2}(\cos 2x - \cos 8x)$
 Since $A - B = 2x$
 $= \frac{1}{2} \times 2 \sin 5x \sin 3x$ and $A + B = 8x$
 $= \sin 5x \sin 3x$ Then $A = 5x$
 and $B = 3x$

$\therefore a = 5 \text{ and } b = 3$ - 1
 or $a = 3 \text{ and } b = 5$

(c) $p + q = -\frac{b}{a} \therefore ap + aq = -b$ --- (1)
 $pq = \frac{c}{a} \therefore apq = c$

$\therefore \frac{q}{ap + b} + \frac{p}{aq + b} = \frac{q}{-aq} + \frac{p}{-ap}$ using (1)
 $= -\frac{1}{a} + \frac{1}{a} = -\frac{2}{a}$

(d) (i) $\int_0^4 f(x) dx = \frac{1}{2} \times 1 \times 1 - \frac{1}{2} (2\frac{1}{2} + 1) \times 1 + \frac{1}{2} \times \frac{1}{2} \times 1$
 $= \frac{1}{2} - 1\frac{3}{4} + \frac{1}{4}$
 $= -1$ - 1

(ii) $\int_0^a f(x) dx$ has the smallest value when it is most negative
 $\therefore a = 3\frac{1}{2}$ - 1